# Lecture 18: Misspecification

Economics 326 — Introduction to Econometrics II

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## Strong exogeneity and the conditional expectation function (CEF)

- Consider the linear regression model

$$Y_i = \beta_0 + \beta_1 X_i + U_i.$$

- When the errors are **strongly exogenous**, i.e. $\mathrm{E}[U_i \mid X_i] = 0$, the linear regression model defines the **CEF** of $Y$ conditional on $X$:

$$\begin{aligned}
\mathrm{CEF}_Y(X_i) &\equiv \mathrm{E}[Y_i \mid X_i] \\
&= \mathrm{E}[\beta_0 + \beta_1 X_i + U_i \mid X_i] \\
&= \beta_0 + \beta_1 X_i + \mathrm{E}[U_i \mid X_i] \\
&= \beta_0 + \beta_1 X_i.
\end{aligned}$$

## Weak exogeneity

$$\begin{aligned}
Y_i &= \beta_0 + \beta_1 X_i + U_i, \\
\mathrm{E}[U_i] &= 0
\end{aligned}$$

- Suppose the errors are only **weakly exogenous**:

$$\mathrm{E}[U_i X_i] = 0.$$

- In this case,

$$\mathrm{CEF}_Y(X_i) \neq \beta_0 + \beta_1 X_i.$$

- **Question:** What does the econometrician estimate when running a linear regression and the regressors are *not* strongly exogenous?

## Linear regression as a misspecified CEF

- Suppose that

$$\mathrm{E}[Y_i \mid X_i] = g(X_i),$$

where $g$ is some unknown *nonlinear* function. Thus, the **true** CEF is $g(X_i) \neq \beta_0 + \beta_1 X_i$.

- Define

$$V_i = Y_i - \mathrm{E}[Y_i \mid X_i],$$

so we can write the true model as

$$\begin{aligned}
Y_i &= g(X_i) + V_i, \\
\mathrm{E}[V_i \mid X_i] &= 0.
\end{aligned}$$

## Linear regression as a misspecified CEF

- Write

$$
\begin{aligned}
Y_i &= g\left(X_i\right) + V_i \\
&= \beta_0 + \beta_1 X_i - \beta_0 - \beta_1 X_i + g\left(X_i\right) + V_i
\end{aligned}
$$

- or

$$Y_i = \beta_0 + \beta_1 X_i + U_i,$$

where

$$U_i = V_i + g\left(X_i\right) - \beta_0 - \beta_1 X_i.$$

- Can we find $\beta_0$ and $\beta_1$ so that $\mathrm{E}\left[U_i\right] = 0$ and $\mathrm{E}\left[X_i U_i\right] = 0$? If yes, how can we interpret such $\beta_0$ and $\beta_1$?

## Linear regression as a misspecified CEF

$$U_i = V_i + g\left(X_i\right) - \beta_0 - \beta_1 X_i.$$

- Note that

$$
\begin{aligned}
\mathrm{E}\left[U_i\right] &= \mathrm{E}\left[V_i + g\left(X_i\right) - \beta_0 - \beta_1 X_i\right] \\
&= \mathrm{E}\left[V_i\right] + \mathrm{E}\left[g\left(X_i\right) - \beta_0 - \beta_1 X_i\right] \\
&= \mathrm{E}\left[g\left(X_i\right) - \beta_0 - \beta_1 X_i\right],
\end{aligned}
$$

- and

$$
\begin{aligned}
\mathrm{E}\left[U_i X_i\right] &= \mathrm{E}\left[\left(V_i + g\left(X_i\right) - \beta_0 - \beta_1 X_i\right) X_i\right] \\
&= \mathrm{E}\left[V_i X_i\right] + \mathrm{E}\left[\left(g\left(X_i\right) - \beta_0 - \beta_1 X_i\right) X_i\right] \\
&= \mathrm{E}\left[\left(g\left(X_i\right) - \beta_0 - \beta_1 X_i\right) X_i\right].
\end{aligned}
$$

- Thus, to have $\mathrm{E}\left[U_i\right] = \mathrm{E}\left[U_i X_i\right] = 0$, we need to find $\beta_0$ and $\beta_1$ such that

$$
\begin{aligned}
\mathrm{E}\left[g\left(X_i\right) - \beta_0 - \beta_1 X_i\right] &= 0, \\
\mathrm{E}\left[\left(g\left(X_i\right) - \beta_0 - \beta_1 X_i\right) X_i\right] &= 0.
\end{aligned}
$$

## Linear approximation of the CEF

- Consider the following approximation problem:

$$\min_{b_0, b_1} \mathrm{E}\left[\left(g\left(X_i\right) - b_0 - b_1 X_i\right)^2\right].$$

- We are approximating the CEF by linear functions.
- Among the linear functions, we are looking for the **best** linear approximation in the **mean squared error (MSE)** sense.

## Linear approximation of the CEF

$$\min_{b_0,b_1} \text{MSE}(b_0,b_1),$$
$$\text{MSE}(b_0,b_1) = \text{E}\left[(g(X_i) - b_0 - b_1X_i)^2\right].$$

- Let $\beta_0$ and $\beta_1$ denote the solution: $(\beta_0,\beta_1) = \arg\min_{b_0,b_1} \text{MSE}(b_0,b_1)$.
- The first-order conditions are:

$$\frac{\partial \text{MSE}(\beta_0,\beta_1)}{\partial b_0} = -2\text{E}[g(X_i) - \beta_0 - \beta_1X_i] = 0.$$
$$\frac{\partial \text{MSE}(\beta_0,\beta_1)}{\partial b_1} = -2\text{E}[(g(X_i) - \beta_0 - \beta_1X_i)X_i] = 0.$$

## Linear regression as the best linear approximation of the CEF

- We have

$$\begin{aligned} Y_i &= \beta_0 + \beta_1X_i + U_i, \\ U_i &= V_i + g(X_i) - \beta_0 - \beta_1X_i. \end{aligned}$$

- With $(\beta_0,\beta_1) = \arg\min_{b_0,b_1} \text{E}\left[(g(X_i) - b_0 - b_1X_i)^2\right]$,

$$\text{E}[U_i] = 0 \text{ and } \text{E}[U_iX_i] = 0.$$

- Thus, the linear regression model gives us the best linear approximation of the CEF (in the MSE sense).

## Misspecification and heteroskedasticity

- We have

$$\begin{aligned} Y_i &= \beta_0 + \beta_1X_i + U_i, \\ U_i &= V_i + g(X_i) - \beta_0 - \beta_1X_i. \end{aligned}$$

- Suppose that the "true" error $V_i$ is homoskedastic: $\text{E}[V_i^2 \mid X_i] = \sigma_V^2$ for all $X_i$.
- $U_i$ is **heteroskedastic** if $g(X_i) \neq \beta_0 + \beta_1X$:

$$\begin{aligned} \text{E}[U_i^2 \mid X_i] &= \text{E}\left[(V_i + g(X_i) - \beta_0 - \beta_1X_i)^2 \mid X_i\right] \\ &= \text{E}\left[V_i^2 + (g(X_i) - \beta_0 - \beta_1X)^2 \mid X_i\right] \\ &\quad + \text{E}[2V_i(g(X_i) - \beta_0 - \beta_1X_i) \mid X_i] \\ &= \sigma_V^2 + (g(X_i) - \beta_0 - \beta_1X)^2. \end{aligned}$$